Kessler Werkzeugmaschinen

KX-500

5-axis machining centre

Operator and Maintenance Manual

Revision 3

For qualified maintenance personnel

1 Safety

1.1 General safety

Only qualified maintenance personnel who have read this manual may service the KX-500. Wear safety glasses and cut-resistant gloves when you handle tools, chips and coolant. Never run the machine with the guarding open or a safety device bypassed. Report damage to the guarding to your supervisor before you continue.

1.2 Lockout and tagout

Before you work inside the guarding or on the spindle, the tool changer or the coolant system, switch off the main isolator and fit your personal lock and tag. Release the compressed air to the tool changer and the drawbar, and wait until the spindle has stopped. Verify zero energy before you start, and remove your lock only when the work is finished.

2 Alarm table

The control shows these codes. Likely causes are listed in order of likelihood.

Code	Message	Severity	Likely causes	Action
SP-001	Warm-up cycle complete	info	none	No action required.
SP-104	Spindle temperature high	critical	worn spindle bearings (most likely) blocked coolant filter (possible)	See 4.1 Replace spindle bearing set (plant ref PROC-CNC-01) See 4.2 Check spindle vibration from the HMI (plant ref PROC-CNC-02) See 4.3 Clean coolant filter and flush lines (plant ref PROC-CNC-03)
SP-117	Spindle vibration above limit	warning	worn spindle bearings (most likely)	See 4.1 Replace spindle bearing set (plant ref PROC-CNC-01) See 4.2 Check spindle vibration from the HMI (plant ref PROC-CNC-02)
CL-210	Coolant pressure low	warning	blocked coolant filter (most likely)	See 4.3 Clean coolant filter and flush lines (plant ref PROC-CNC-03)
CL-214	Coolant flow interrupted	critical	blocked coolant filter (most likely)	See 4.3 Clean coolant filter and flush lines (plant ref PROC-CNC-03)
TC-305	Tool clamp not confirmed	critical	weak tool clamp springs or faulty clamp sensor (most likely)	See 4.4 Replace tool clamp disc springs (plant ref PROC-CNC-04)
TC-311	Tool change timeout	warning	weak tool clamp springs or faulty clamp sensor (most likely)	See 4.4 Replace tool clamp disc springs (plant ref PROC-CNC-04)

3 Troubleshooting table

Find the symptom, then follow the remedy.

Symptom	Probable cause	Remedy
spindle running hot and noisy at high rpm	worn spindle bearings	See 4.1 Replace spindle bearing set (plant ref PROC-CNC-01) See 4.2 Check spindle vibration from the HMI (plant ref PROC-CNC-02)
coolant pressure dropping, parts coming out hot	blocked coolant filter	See 4.3 Clean coolant filter and flush lines (plant ref PROC-CNC-03)
tool not clamping after change, machine stops	weak tool clamp springs or faulty clamp sensor	See 4.4 Replace tool clamp disc springs (plant ref PROC-CNC-04)

4 Procedures

4.1 Replace spindle bearing set

Plant ref: PROC-CNC-01 | Lockout required: yes | Time: 240 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Stop the machine, apply lockout and tagout on the main isolator, and verify zero energy.
2. Remove the spindle cover and disconnect the drawbar and coolant lines.
3. Extract the spindle cartridge with the puller and mount it on the bench.
4. Press out both bearings and fit the new BRG-7014-ACD pair with the spacer.
5. Fit a new SEAL-SPDL-70 front seal and pack the bearings with GREASE-KLB-2.
6. Refit the cartridge, run the 20-minute warm-up cycle and check that vibration is below 2.5 mm/s.

Order no.	Description	Qty
BRG-7014-ACD	Spindle angular contact bearing 7014 ACD	2
SEAL-SPDL-70	Spindle front seal 70 mm	1
GREASE-KLB-2	Spindle bearing grease 400 g	1

4.2 Check spindle vibration from the HMI

Plant ref: PROC-CNC-02 | Lockout required: no | Time: 20 min

No isolation needed. Work from the HMI only.

1. Open Diagnostics, then Spindle, on the HMI and start the vibration test at 6000 rpm.
2. Record the vibration reading and the bearing temperature.
3. If vibration is above 2.5 mm/s or the temperature is above 65 C, plan PROC-CNC-01.
4. Log the readings in the shift handover.

Parts required: none.

4.3 Clean coolant filter and flush lines

Plant ref: PROC-CNC-03 | Lockout required: yes | Time: 45 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Stop the coolant pump and apply lockout and tagout on the coolant pump breaker.
2. Drain the filter housing into the waste tray.
3. Replace the filter element with FLT-CL-25 and clean the housing.
4. Flush the lines at low pressure for 5 minutes and check for leaks.
5. Remove the lockout, restart the pump and confirm pressure above 18 bar.

Order no.	Description	Qty
FLT-CL-25	Coolant filter element 25 micron	1

4.4 Replace tool clamp disc springs

Plant ref: PROC-CNC-04 | Lockout required: yes | Time: 120 min

WARNING: Apply lockout and tagout before starting. Verify zero energy.

1. Stop the machine, apply lockout and tagout, and release air pressure to the tool changer.
2. Remove the drawbar assembly from the spindle.
3. Replace the disc spring stack with SPR-TC-40.
4. Replace the clamp confirmation sensor SENS-PROX-08 and set its gap to 0.8 mm.
5. Refit the drawbar and confirm a clamping force above 18 kN with the test gauge.

Order no.	Description	Qty
SPR-TC-40	Tool clamp disc spring set	1
SENS-PROX-08	Inductive proximity sensor M8	1

5 Spare parts list

Order spare parts by part number.

Part number	Description	Fits
BRG-7014-ACD	Spindle angular contact bearing 7014 ACD	Spindle
CLEANER-CL-5L	Coolant system cleaner, 5 litre	Coolant system
COLLET-ER32	Collet set ER32	Spindle
FLT-CL-25	Coolant filter element 25 micron	Coolant system
GREASE-KLB-2	Spindle bearing grease 400 g	Spindle
HOLDER-BT40	Tool holder BT40	Tool changer
NOZZLE-CL-06	Coolant nozzle 6 mm	Coolant system
PUMP-CL-04	Coolant pump 0.4 kW	Coolant system
SEAL-SPDL-70	Spindle front seal 70 mm	Spindle
SENS-PROX-08	Inductive proximity sensor M8	Tool changer
SENS-PT100-01	Temperature sensor PT100	Spindle
SPR-TC-40	Tool clamp disc spring set	Tool changer